

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5371

5 Credits: 4

Program Core

Source-grounded

Microcontroller and Embedded

□ Acquire knowledge about microcontrollers embedded processors and their

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5371 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5371.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Integrated Circuit Design and Fabrication
Course code	5371
Course title	Microcontroller and Embedded
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

21 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ Acquire knowledge about microcontrollers embedded processors and their applications.
- □ Introduce the importance of embedded systems
- □ Introduce ATMEGA 32 microcontroller and its suitability for embedded systems
- Familiarize embedded C programming
- □ Familiarize serial communication and data transfer between embedded system

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding architecture Explain the architecture and functionality of 8051	See official syllabus
CO2	14 Understanding microcontroller Make use of AVR Microcontrollers to develop	See official syllabus
CO3	15 Applying embedded programs using embedded C Make use of AVR microcontroller to interface	See official syllabus
CO4	various peripheral devices using embedded and 14 Applying Familiarize RTOS	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 3 3

Row	Extracted official mapping
CO4	CO4 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the basics of embedded systems and its architecture	5	As specified
Module 2	Explain the architecture and functionality of 8051 microcontroller	4	As specified
Module 3	embedded C	6	As specified
Module 4	using embedded C.	6	As specified

MODULE 1

Explain the basics of embedded systems and its architecture

This module is presented from the official syllabus scope for Microcontroller and Embedded. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the basics of embedded systems and its architecture
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Understanding purpose computer 1 Classify embedded systems and explain

Understanding purpose computer 1 Classify embedded systems and explain is an official topic in Module 1 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding purpose computer 1 Classify embedded systems and explain using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding purpose computer 1 classify embedded systems and explain should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Understanding purpose computer 1 Classify embedded systems and explain means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എന്നുൾപ്പെടെയുള്ള ഉപയോക്താക്കൾക്ക് ഉപയോഗിക്കാവുന്ന വിവിധ ഉപകരണങ്ങൾ, steps, application എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ ഉപയോഗിക്കാവുന്നവർ.

▼ Understanding its application areas 2 Explain Hardware Architecture of embedded 6

Understanding its application areas 2 Explain Hardware Architecture of embedded 6 is an official topic in Module 1 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding its application areas 2 Explain Hardware Architecture of embedded 6 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding its application areas 2 explain hardware architecture of embedded 6 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Understanding its application areas 2 Explain Hardware Architecture of embedded 6 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Understanding its application areas 2 Explain Hardware Architecture of embedded 6 definition/meaning , steps, application Technical terms English-

▼ Understanding system Explain Software Architecture of embedded 4

Understanding system Explain Software Architecture of embedded 4 is an official topic in Module 1 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding system Explain Software Architecture of embedded 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding system explain software architecture of embedded 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Understanding system Explain Software Architecture of embedded 4 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Understanding system Explain Software Architecture of embedded 4 definition/meaning. steps, application. Technical terms English-.

▼ Understanding system

Understanding system is an official topic in Module 1 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding system means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Understanding system
 definition/meaning . steps, application . Technical terms English-
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▼ Explain Real Time Operating Systems 2 Understanding Embedded Systems - Definition, difference from general purpose computers - Classification of embedded systems, Application areas. Hardware Architecture of embedded system - Building blocks of an embedded system, Core of embedded system - categories, Memory -ROM and RAM, Sensors and I/O subsystems (LED, Relay, Stepper motor). On board (I2C, SPI, UART) and external communication interfaces (RS 232, USB, Bluetooth, Wi-Fi). Software Architecture of embedded system- Software Layers. Kernel, types of operating systems- GPOS, RTOS. Real time operating systems- Tasks, process, threads, multiprocessing and multi-tasking List popular Real Time Operating Systems (any 10)

Explain Real Time Operating Systems 2 Understanding Embedded Systems - Definition, difference from general purpose computers - Classification of embedded systems, Application areas. Hardware Architecture of embedded system - Building blocks of an embedded system, Core of embedded system - categories, Memory -ROM and RAM, Sensors and I/O subsystems (LED, Relay, Stepper motor). On board (I2C, SPI, UART) and external communication interfaces (RS 232, USB, Bluetooth, Wi-Fi). Software Architecture of embedded system-Software Layers. Kernel, types of operating systems- GPOS, RTOS. Real time operating systems- Tasks, process, threads, multiprocessing and multi-tasking List popular Real Time Operating Systems (any 10) is an official topic in Module 1 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Real Time Operating Systems 2 Understanding Embedded Systems - Definition, difference from general purpose computers - Classification of embedded systems, Application areas. Hardware Architecture of embedded system – Building blocks of an embedded system, Core of embedded system – categories, Memory –ROM and RAM, Sensors and I/O subsystems (LED, Relay, Stepper motor). On board (I2C, SPI, UART) and external communication interfaces (RS 232, USB, Bluetooth, Wi-Fi). Software Architecture of embedded system-Software Layers. Kernel, types of operating systems- GPOS, RTOS. Real time operating systems- Tasks, process, threads, multiprocessing and multi-tasking List popular Real Time Operating Systems (any 10) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain real time operating systems 2 understanding embedded systems - definition, difference from general purpose computers - classification of embedded systems, application areas. hardware architecture of embedded system – building blocks of an embedded system, core of embedded system – categories, memory –rom and ram, sensors and i/o subsystems (led, relay, stepper motor). on board (i2c, spi, uart) and external communication interfaces (rs 232, usb, bluetooth, wi-fi). software architecture of embedded system-software layers. kernel, types of operating systems- gpos, rtos. real time operating systems- tasks, process, threads, multiprocessing and multi-tasking list popular real time operating systems (any 10) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain Real Time Operating Systems 2 Understanding Embedded Systems - Definition, difference from general purpose computers - Classification of embedded systems, Application areas. Hardware Architecture of embedded system – Building blocks of an embedded system, Core of embedded system – categories, Memory –ROM and RAM, Sensors and I/O subsystems (LED, Relay, Stepper motor). On board (I2C, SPI, UART) and external communication interfaces (RS 232, USB, Bluetooth, Wi-Fi). Software Architecture of embedded system- Software Layers. Kernel, types of operating systems- GPOS, RTOS. Real time operating systems- Tasks, process, threads, multiprocessing and multi-tasking List popular Real Time Operating Systems (any 10) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain Real Time Operating Systems 2 Understanding Embedded Systems - Definition, difference from general purpose computers - Classification of embedded systems, Application areas. Hardware Architecture of embedded system – Building blocks of an embedded system, Core of embedded system – categories, Memory –ROM and RAM, Sensors and I/O subsystems (LED, Relay, Stepper motor). On board (I2C, SPI, UART) and external communication interfaces (RS 232, USB, Bluetooth, Wi-Fi). Software Architecture of embedded system-Software Layers. Kernel, types of operating systems- GPOS, RTOS. Real time operating systems- Tasks, process, threads, multiprocessing and multi-tasking List popular Real Time Operating Systems (any 10)

definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-പദപരിഭാഷണം പദപരിഭാഷണം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the architecture and functionality of 8051 microcontroller

This module is presented from the official syllabus scope for Microcontroller and Embedded. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the architecture and functionality of 8051 microcontroller
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ List the features of 8051 microcontroller. 2 Understanding Explain the block diagram of 8051

List the features of 8051 microcontroller. 2 Understanding Explain the block diagram of 8051 is an official topic in Module 2 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List the features of 8051 microcontroller. 2 Understanding Explain the block diagram of 8051 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list the features of 8051 microcontroller. 2 understanding explain the block diagram of 8051 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What List the features of 8051 microcontroller. 2 Understanding Explain the block diagram of 8051 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ List the features of 8051 microcontroller. 2 Understanding Explain the block diagram of 8051 □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

▼ 2 Understanding microcontroller

2 Understanding microcontroller is an official topic in Module 2 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding microcontroller using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding microcontroller should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding microcontroller means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding microcontroller
 പരീക്ഷയ്ക്കായി definition/meaning നൽകണം. പരീക്ഷയിൽ പരീക്ഷണ
 പരീക്ഷണം, steps, application നൽകണം പരീക്ഷണ പരീക്ഷണം. Technical terms
 English-ൽ പരീക്ഷ പരീക്ഷണം.

▼ Explain memory organization of 8051. 6 Understanding Explain concept of interrupts, types, priority,

Explain memory organization of 8051. 6 Understanding Explain concept of interrupts, types, priority, is an official topic in Module 2 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain memory organization of 8051. 6 Understanding Explain concept of interrupts, types, priority, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain memory organization of 8051. 6 understanding explain concept of interrupts, types, priority, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain memory organization of 8051. 6 Understanding Explain concept of interrupts, types, priority, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain memory organization of 8051. 6 Understanding Explain concept of interrupts, types, priority, definition/meaning. definition/meaning, steps, application. Technical terms English- Malayalam.

▼ 4 Understanding registers for interrupts and handling of interrupts.

Introduction to Microcontroller, simple block diagram of 8051, Data memory organization, program memory organization, PSW register.

8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register.

Make use of AVR Microcontrollers to develop embedded programs using

4 Understanding registers for interrupts and handling of interrupts.

Introduction to Microcontroller, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Make use of AVR Microcontrollers to develop embedded programs using is an official topic in Module 2 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding registers for interrupts and handling of interrupts. Introduction to Microcontroller, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Make use of AVR Microcontrollers to develop embedded programs using using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding registers for interrupts and handling of interrupts. introduction to microcontroller, simple block diagram of 8051, data memory organization, program memory organization, psw register. 8051 interrupts: concept, types, interrupt service routine (isr), ie register, steps in executing an interrupt, interrupt priority, ip register. make use of avr microcontrollers to develop embedded programs using should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding registers for interrupts and handling of interrupts. Introduction to Microcontroller, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Make use of AVR Microcontrollers to develop embedded programs using means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

► **Official syllabus detail and terminology (expand in digital version)**

<http://127.0.0.1:9876/revision-2026-content/lessons/lessons-5371.html?autoPrintNotes=1>

MODULE 3**embedded C**

This module is presented from the official syllabus scope for Microcontroller and Embedded. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: embedded C
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding criteria to select a microcontroller.

2 Understanding criteria to select a microcontroller. is an official topic in Module 3 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding criteria to select a microcontroller. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding criteria to select a microcontroller. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding criteria to select a microcontroller. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding criteria to select a microcontroller. ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം, steps, application ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English- ഫലപ്രസാദം ഫലപ്രസാദം.

▼ Explain block diagram of Atmega32. 3 Understanding Illustrate Registers, Memory organization, Status

Explain block diagram of Atmega32. 3 Understanding Illustrate Registers, Memory organization, Status is an official topic in Module 3 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain block diagram of Atmega32. 3 Understanding Illustrate Registers, Memory organization, Status using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain block diagram of atmega32. 3 understanding illustrate registers, memory organization, status should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain block diagram of Atmega32. 3 Understanding Illustrate Registers, Memory organization, Status means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain block diagram of Atmega32. 3 Understanding Illustrate Registers, Memory organization, Status
 പരീക്ഷയിൽ എഴുതേണ്ടത് definition/meaning പരീക്ഷയിൽ ഉൾപ്പെടുന്നു. പരീക്ഷയിൽ ഉൾപ്പെടുന്ന
 പരീക്ഷയിൽ, steps, application പരീക്ഷയിൽ ഉൾപ്പെടുന്നു. Technical terms
 English-ൽ ഉൾപ്പെടുന്നു പരീക്ഷയിൽ ഉൾപ്പെടുന്നു.

▼ register, Program counter, I/O ports and its registers, 3 Understanding Interrupts, priority of interrupts

register, Program counter, I/O ports and its registers, 3 Understanding Interrupts, priority of interrupts is an official topic in Module 3 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify register, Program counter, I/O ports and its registers, 3 Understanding Interrupts, priority of interrupts using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, register, program counter, i/o ports and its registers, 3 understanding interrupts, priority of interrupts should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What register, Program counter, I/O ports and its registers, 3 Understanding Interrupts, priority of interrupts means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-റജിസ്റ്റർ, Program counter, I/O ports and its registers, 3 Understanding Interrupts, priority of interrupts ഏതാളെങ്കിലും ഏതാളെങ്കിലും definition/meaning ഏതാളെങ്കിലും. ഏതാളെങ്കിലും ഏതാളെങ്കിലും steps, application ഏതാളെങ്കിലും ഏതാളെങ്കിലും. Technical terms English-ൽ ഏതാളെങ്കിലും.

▼ Illustrate Timers in AVR 3 Understanding Develop embedded C programs for logic

Illustrate Timers in AVR 3 Understanding Develop embedded C programs for logic is an official topic in Module 3 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Timers in AVR 3 Understanding Develop embedded C programs for logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate timers in avr 3 understanding develop embedded c programs for logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Illustrate Timers in AVR 3 Understanding Develop embedded C programs for logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Illustrate Timers in AVR 3 Understanding Develop embedded C programs for logic definition/meaning . . . , steps, application Technical terms English-

▼ 3 Applying operations, data conversions and I/O operations.

3 Applying operations, data conversions and I/O operations. is an official topic in Module 3 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying operations, data conversions and I/O operations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying operations, data conversions and i/o operations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Applying operations, data conversions and I/O operations. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Applying operations, data conversions and I/O operations. definition/meaning . steps, application . Technical terms English- .

▼ **Compare 8051 and Atmega 32 Microcontroller. 1 Understanding AVR Microcontroller Architecture - Comparison of AVR family members and Selection of a microcontroller, ATmega32- Simplified Block diagram of ATmega32 microcontroller - Registers, - data memory - I/O memory - SFRs - internal data SRAM, Status register, Program Counter and Program ROM space, I/O ports, Registers associated with I/O ports, Timers- 0,1,2 (Block level) associated registers, Interrupts. AVR programming using embedded C-Data types, I/O programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2. Comparison of 8051 and Atmega 32 Microcontroller. Make use of AVR microcontroller to interface various peripheral devices**

Compare 8051 and Atmega 32 Microcontroller. 1 Understanding AVR Microcontroller Architecture - Comparison of AVR family members and Selection of a microcontroller, ATmega32- Simplified Block diagram of ATmega32 microcontroller - Registers, - data memory - I/O memory -SFRs - internal data SRAM, Status register, Program Counter and Program ROM space, I/O ports, Registers associated with I/O ports, Timers- 0,1,2 (Block level) associated registers, Interrupts. AVR programming using embedded C-Data types, I/O programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2. Comparison of 8051 and Atmega 32 Microcontroller. Make use of AVR microcontroller to interface various peripheral devices is an official topic in Module 3 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare 8051 and Atmega 32 Microcontroller. 1 Understanding AVR Microcontroller Architecture - Comparison of AVR family members and Selection of a microcontroller, ATmega32- Simplified Block diagram of ATmega32 microcontroller - Registers, - data memory - I/O memory -SFRs - internal data SRAM, Status register, Program Counter and Program ROM space, I/O ports, Registers associated with I/O ports, Timers- 0,1,2 (Block level) associated registers, Interrupts. AVR programming using embedded C-Data types, I/O programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2. Comparison of 8051 and Atmega 32 Microcontroller. Make use of AVR microcontroller to interface various peripheral devices using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare 8051 and atmega 32 microcontroller. 1 understanding avr microcontroller architecture - comparison of avr family members and selection of a microcontroller, atmega32- simplified block diagram of atmega32 microcontroller - registers, - data memory - i/o memory -sfrs - internal data sram, status register, program counter and program rom space, i/o ports, registers associated with i/o ports, timers- 0,1,2 (block level) associated registers, interrupts. avr programming using embedded c-data types, i/o programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2. comparison of 8051 and atmega 32 microcontroller. make use of avr microcontroller to interface various peripheral devices should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare 8051 and Atmega 32 Microcontroller. 1 Understanding AVR Microcontroller Architecture - Comparison of AVR family members and Selection of a microcontroller, ATmega32- Simplified Block diagram of ATmega32 microcontroller - Registers, - data memory - I/O memory - SFRs - internal data SRAM, Status register, Program Counter and Program ROM space, I/O ports, Registers associated with I/O ports, Timers- 0,1,2 (Block level) associated registers, Interrupts. AVR programming using embedded C-Data types, I/O programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2. Comparison of 8051 and Atmega 32 Microcontroller. Make use of AVR microcontroller to interface various peripheral devices means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Compare 8051 and Atmega 32 Microcontroller. 1 Understanding AVR Microcontroller Architecture - Comparison of AVR family members and Selection of a microcontroller, ATmega32- Simplified Block diagram of ATmega32 microcontroller - Registers, - data memory - I/O memory -SFRs - internal data SRAM, Status register, Program Counter and Program ROM space, I/O ports, Registers associated with I/O ports, Timers- 0,1,2 (Block level) associated registers, Interrupts. AVR programming using embedded C-Data types, I/O programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2. Comparison of 8051 and Atmega 32 Microcontroller. Make use of AVR microcontroller to interface various

peripheral devices ഘടനാപരമായ വിവരണം definition/meaning ഘടനാപരമായ വിവരണം.
ഘടനാപരമായ വിവരണം, steps, application ഘടനാപരമായ വിവരണം ഘടനാപരമായ വിവരണം.
Technical terms English-ഘടനാപരമായ വിവരണം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4**using embedded C.**

This module is presented from the official syllabus scope for Microcontroller and Embedded. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: using embedded C.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding interfacing devices. Illustrate the interfacing of LED, Push button,

1 Understanding interfacing devices. Illustrate the interfacing of LED, Push button, is an official topic in Module 4 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding interfacing devices. Illustrate the interfacing of LED, Push button, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding interfacing devices. illustrate the interfacing of led, push button, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding interfacing devices. Illustrate the interfacing of LED, Push button, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-1 Understanding interfacing devices. Illustrate the interfacing of LED, Push button, definition/meaning. steps, application. Technical terms English- Malayalam.

▼ Relay, Sensors and Seven segment Display with 3 Applying AVR

Relay, Sensors and Seven segment Display with 3 Applying AVR is an official topic in Module 4 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Relay, Sensors and Seven segment Display with 3 Applying AVR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, relay, sensors and seven segment display with 3 applying avr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Relay, Sensors and Seven segment Display with 3 Applying AVR means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-റിലേ, സെൻസറുകളും ഏഴ് സെഗ്മെന്റ് ഡിസ്പ്ലൈ വുമു 3 എപ്ലൈംഗ് AVR ഡെഫിനിഷൻ/മിനിംഗ് ഡെഫിനിഷൻ. ഡെഫിനിഷൻ ഡെഫിനിഷൻ, steps, application ഡെഫിനിഷൻ ഡെഫിനിഷൻ. Technical terms English-റിലേ ഡെഫിനിഷൻ ഡെഫിനിഷൻ.

▼ Make use of AVR to interface LCD and Keyboard 3 Applying Make use of AVR microcontroller to interface DC

Make use of AVR to interface LCD and Keyboard 3 Applying Make use of AVR microcontroller to interface DC is an official topic in Module 4 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Make use of AVR to interface LCD and Keyboard 3 Applying Make use of AVR microcontroller to interface DC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, make use of avr to interface lcd and keyboard 3 applying make use of avr microcontroller to interface dc should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Make use of AVR to interface LCD and Keyboard 3 Applying Make use of AVR microcontroller to interface DC means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Make use of AVR to interface LCD and Keyboard 3 Applying Make use of AVR microcontroller to interface DC
 പരീക്ഷയിൽ എഴുതേണ്ടത് definition/meaning ഉൾപ്പെടെയുള്ളവയാണ്. പരീക്ഷയിൽ എഴുതേണ്ടത്
 principle, steps, application ഉൾപ്പെടെയുള്ളവയാണ്. Technical terms
 English-ൽ എഴുതേണ്ടത് ഉൾപ്പെടെയുള്ളവയാണ്.

▼ 3 Applying motor, Servo motor and stepper motor and ADC

3 Applying motor, Servo motor and stepper motor and ADC is an official topic in Module 4 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying motor, Servo motor and stepper motor and ADC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying motor, servo motor and stepper motor and adc should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying motor, Servo motor and stepper motor and ADC means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 3 Applying motor, Servo motor and stepper motor and ADC ഓരോന്നിനും definition/meaning ഉൾപ്പെടെ. ഓരോന്നും ഓരോന്നും, steps, application ഉൾപ്പെടെ ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ ഉൾപ്പെടെ.

▼ Implement I2C interfacing with AVR

Implement I2C interfacing with AVR is an official topic in Module 4 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Implement I2C interfacing with AVR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, implement i2c interfacing with avr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Implement I2C interfacing with AVR means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇംഗ്ലീഷ് Implement I2C interfacing with AVR
 ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്
 ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms
 English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

▼ **Understand SPI interface with AVR** 1 **Understanding Interfacing of LED, Push button, Relay, Sensors (Temperature sensor, IR sensor) Seven segment Display, LCD, Keyboard Interfacing, Motor (DC, Servo, Stepper), ADC interfacing. On-board communication interfaces with AVR - basics of I2C interfacing (like real time clock interfacing) and SPI interfacing with AVR** Text/Reference T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 T1 Microcontroller and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Rajkamal, Embedded Systems Architecture, Programming and Design, TMH, T2 2003 K.V. Shibu, Introduction to Embedded Systems, 2e, McGraw Hill Education T3 India, 2016. T4 KVK K Prasad R1 Michael J. Pont, Embedded C, Pearson Education, Second Edition Online resources: Sl.No. Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com> <https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-4-beginners/> 5 https://www.tutorialspoint.com/embedded_systems/index.htm 6 <https://embeddedschool.in/avr-microcontroller-programming/> 7 <https://learn.sparkfun.com/tutorials/i2c/all> 8 <https://www.best-microcontroller-projects.com/i2c-tutorial.html> 9 <https://www.electronicwings.com/avr-atmega/atmega1632-i2c> 10 <https://www.electronicwings.com/avr-atmega/atmega1632-spi> 11 <https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.htm>

Understand SPI interface with AVR 1 Understanding Interfacing of LED, Push button, Relay, Sensors (Temperature sensor, IR sensor) Seven segment Display, LCD, Keyboard Interfacing, Motor (DC, Servo, Stepper), ADC interfacing. On-board communication interfaces with AVR - basics of I2C interfacing (like real time clock interfacing) and SPI interfacing with AVR Text/Reference T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 T1 Microcontroller and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Rajkamal, Embedded Systems Architecture, Programming and Design, TMH, T2 2003

K.V. Shibu, Introduction to Embedded Systems, 2e, McGraw Hill Education T3 India, 2016. T4 KVK K Prasad R1 Michael J. Pont, Embedded C, Pearson Education, Second Edition Online resources: Sl.No. Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com> <https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-4-beginners/> 5 https://www.tutorialspoint.com/embedded_systems/index.htm 6 <https://embeddedschool.in/avr-microcontroller-programming/> 7 <https://learn.sparkfun.com/tutorials/i2c/all> 8 <https://www.best-microcontroller-projects.com/i2c-tutorial.html> 9 <https://www.electronicwings.com/avr-atmega/atmega1632-i2c> 10 <https://www.electronicwings.com/avr-atmega/atmega1632-spi> 11 <https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.htm> is an official topic in Module 4 of Microcontroller and Embedded. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand SPI interface with AVR 1 Understanding Interfacing of LED, Push button, Relay, Sensors (Temperature sensor, IR sensor) Seven segment Display, LCD, Keyboard Interfacing, Motor (DC, Servo, Stepper), ADC interfacing. On-board communication interfaces with AVR – basics of I2C interfacing (like real time clock interfacing) and SPI interfacing with AVR Text/Reference T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 T1 Microcontroller and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Rajkamal, Embedded Systems Architecture, Programming and

Design, TMH, T2 2003 K.V. Shibu, Introduction to Embedded Systems, 2e, McGraw Hill Education T3 India, 2016. T4 KVK K Prasad R1 Michael J. Pont, Embedded C, Pearson Education, Second Edition Online resources: Sl.No. Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com> <https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-4-beginners/> 5 https://www.tutorialspoint.com/embedded_systems/index.htm 6 <https://embeddedschool.in/avr-microcontroller-programming/> 7 <https://learn.sparkfun.com/tutorials/i2c/all> 8 <https://www.best-microcontroller-projects.com/i2c-tutorial.html> 9 <https://www.electronicwings.com/avr-atmega/atmega1632-i2c> 10 <https://www.electronicwings.com/avr-atmega/atmega1632-spi> 11 <https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.htm> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand spi interface with avr 1 understanding interfacing of led, push button, relay, sensors (temperature sensor, ir sensor) seven segment display, lcd, keyboard interfacing, motor (dc, servo, stepper), adc interfacing. on-board communication interfaces with avr – basics of i2c interfacing (like real time clock interfacing) and spi interfacing with avr text/reference t/r book title/author muhammad ali mazidi and janice gillispi mazidi - the 8051 t1 microcontroller and embedded systems using assembly and c - pearson education- second edition. rajkamal, embedded systems architecture, programming and design, tmh, t2 2003 k.v. shibu, introduction to embedded systems, 2e, mcgraw hill education t3 india, 2016. t4 kvk k prasad r1 michael j. pont, embedded c, pearson education, second edition online resources: sl.no. website link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com>

<https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-4-beginners/> 5 https://www.tutorialspoint.com/embedded_systems/index.htm 6 <https://embeddedschool.in/avr-microcontroller-programming/> 7 <https://learn.sparkfun.com/tutorials/i2c/all> 8 <https://www.best-microcontroller-projects.com/i2c-tutorial.html> 9 <https://www.electronicwings.com/avr-atmega/atmega1632-i2c> 10 <https://www.electronicwings.com/avr-atmega/atmega1632-spi> 11 <https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.htm> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand SPI interface with AVR 1 Understanding Interfacing of LED, Push button, Relay, Sensors (Temperature sensor, IR sensor) Seven segment Display, LCD, Keyboard Interfacing, Motor (DC, Servo, Stepper), ADC interfacing. On-board communication interfaces with AVR – basics of I2C interfacing (like real time clock interfacing) and SPI interfacing with AVR Text/Reference T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 T1 Microcontroller and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Rajkamal, Embedded Systems Architecture, Programming and Design, TMH, T2 2003 K.V. Shibu, Introduction to Embedded Systems, 2e, McGraw Hill Education T3 India, 2016. T4 KVK K Prasad R1 Michael J. Pont, Embedded C, Pearson Education, Second Edition Online resources: Sl.No. Website Link 1 https://www.electronicshub.org 2 https://www.elprocus.com 3 https://www.tutorialspoint.com https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-4-beginners/ 5 https://www.tutorialspoint.com/embedded_systems/index.htm 6 https://embeddedschool.in/avr-microcontroller-programming/ 7 https://learn.sparkfun.com/tutorials/i2c/all 8 https://www.best-microcontroller-projects.com/i2c-tutorial.html 9 https://www.electronicwings.com/avr-atmega/atmega1632-i2c 10 https://www.electronicwings.com/avr-atmega/atmega1632-spi 11 https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.htm means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Understand SPI interface with AVR 1 Understanding Interfacing of LED, Push button, Relay, Sensors (Temperature sensor, IR sensor) Seven segment Display, LCD, Keyboard Interfacing, Motor (DC, Servo, Stepper), ADC interfacing. On-board communication interfaces with AVR – basics of I2C interfacing (like real time clock interfacing) and SPI interfacing with AVR Text/Reference T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 T1 Microcontroller and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Rajkamal, Embedded Systems Architecture, Programming and Design, TMH, T2 2003 K.V. Shibu, Introduction to Embedded Systems, 2e, McGraw Hill Education T3 India, 2016. T4 KVK K Prasad R1 Michael J. Pont, Embedded C, Pearson Education, Second Edition Online resources: Sl.No. Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com> <https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-4-beginners/> 5 https://www.tutorialspoint.com/embedded_systems/index.htm 6 <https://embeddedschool.in/avr-microcontroller-programming/> 7 <https://learn.sparkfun.com/tutorials/i2c/all> 8 <https://www.best-microcontroller-projects.com/i2c-tutorial.html> 9 <https://www.electronicwings.com/avr-atmega/atmega1632-i2c> 10 <https://www.electronicwings.com/avr-atmega/atmega1632-spi> 11

<https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.htm> നിർവ്വചനം/അർത്ഥം, steps, application നൽകുന്ന വിവരങ്ങൾ നൽകുന്നു. Technical terms English-ൽ നൽകുന്നു.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain Understanding purpose computer 1 Classify embedded systems and explain. (3 marks)

► Define or explain Understanding its application areas 2 Explain Hardware Architecture of embedded 6. (3 marks)

► Define or explain Understanding system Explain Software Architecture of embedded 4. (3 marks)

► Define or explain Understanding system. (3 marks)

Module 2

► Define or explain List the features of 8051 microcontroller. 2 Understanding Explain the block diagram of 8051. (3 marks)

► Define or explain 2 Understanding microcontroller. (3 marks)

► Define or explain Explain memory organization of 8051. 6 Understanding Explain concept of interrupts, types, priority,. (3 marks)

► Define or explain 4 Understanding registers for interrupts and handling of interrupts. Introduction to Microcontroller, simple block diagram of 8051, Data memory organization, program memory

organization, PSW register. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Make use of AVR Microcontrollers to develop embedded programs using. (3 marks)

Module 3

► Define or explain 2 Understanding criteria to select a microcontroller.. (3 marks)

► Define or explain Explain block diagram of Atmega32. 3 Understanding Illustrate Registers, Memory organization, Status. (3 marks)

► Define or explain register, Program counter, I/O ports and its registers, 3 Understanding Interrupts, priority of interrupts. (3 marks)

► Define or explain Illustrate Timers in AVR 3 Understanding Develop embedded C programs for logic. (3 marks)

Module 4

► Define or explain 1 Understanding interfacing devices. Illustrate the interfacing of LED, Push button,. (3 marks)

► Define or explain Relay, Sensors and Seven segment Display with 3 Applying AVR. (3 marks)

► Define or explain Make use of AVR to interface LCD and Keyboard 3 Applying Make use of AVR microcontroller to interface DC. (3 marks)

► Define or explain 3 Applying motor, Servo motor and stepper motor and ADC. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Understanding purpose computer 1 Classify embedded systems and explain.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **List the features of 8051 microcontroller. 2 Understanding Explain the block diagram of 8051.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding criteria to select a microcontroller.**

3-mark explanation: Explain one principle, process or comparison from

Module 4

1-mark definition: State the meaning of **1 Understanding interfacing devices. Illustrate the interfacing of LED, Push button,**

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No. Website Link <https://www.electronicshub.org> <https://www.elprocus.com>
<https://www.tutorialspoint.com>
- <https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-beginners/>
- https://www.tutorialspoint.com/embedded_systems/index.htm
- <https://embeddedschool.in/avr-microcontroller-programming/>
<https://learn.sparkfun.com/tutorials/i2c/all>
- <https://www.best-microcontroller-projects.com/i2c-tutorial.html>
- <https://www.electronicwings.com/avr-atmega/atmega1632-i2c>
- <https://www.electronicwings.com/avr-atmega/atmega1632-spi>
- <https://www.intel.com/content/www/us/en/docs/programmable/683525/21-/overview-of-the-microc-os-ii-rtos.htm>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5371. Verify institutional updates against the current official curriculum.